

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	P. Dada Kalander	Reg. No.:	192424272
Section / Batch:	B. Tech AI&DS (2024 Batch)	Date:	17-08-2026

Code of Conduct

I P. Dada Kalander (192424272) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Dada Kalander

Instructions

- Read each question carefully before selecting your answer.
- This assessment consists of 50 multiple-choice questions.
- Do not use external resources, AI tools, textbooks, or assistance from others during the assessment unless explicitly permitted by the instructor.
- Upload Score Card in GCR and submit score card printout.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Context-Free Grammars (CFG)	CFG components, productions, terminals, non-terminals, recursion, sentence generation	1 – 10
Sentence Construction and Agreement	Parse trees, sentence structure, grammatical agreement, syntactic validation	11 – 20
Subcategorization	Verb frames, complement requirements, argument structure analysis	21 – 25
Top-Down Parsing	Parsing strategy, derivation process, recursion issues, grammar expansion	26 – 30
Earley Parsing	Predictor, Scanner, Completer operations, ambiguity handling	31 – 35
Feature Structures	Feature representation, unification, constraint checking, agreement enforcement	36 – 40
Probabilistic Context-Free Grammars (PCFG)	Rule probabilities, ambiguity resolution, probabilistic parsing	41 – 48
Integrated Syntax Analysis	CFG + Feature Structures + PCFG integration in NLP applications	49 – 50

Annexure A – MCQ Test

<https://forms.gle/WSv28zfaU5B2aWcy6>

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Number of Correct Answers	100%	45 to 50 correct answers	38 to 44 correct answers	25 to 37 correct answers	0 to 24 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

No. of Questions	Number of Correct Answers	Accuracy	Total %
50			

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

14/08/2026

MCA Test

DSA0304 - Natural Language Processing

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Name: P Dada kalander

Reg No: P2424272

- ① [B] Recursive production rules allow hierarchical sentence generation. ✓
- ② [C] Verb Sub-Categorization. ✓
- ③ [C] Finite state Automation. ✗
- ④ [A] morphological analyzer. ✗
- ⑤ [B] Left recursion causing infinite expansion. ✓
- ⑥ [B] Uses Predictor, Scanner, & completer operators to process partial input. ✓
- ⑦ [A] Regular expression matching. ✗
- ⑧ [A] Chooses the parse tree with the max rule probability. ✓
- ⑨ [C] Finite state Transducer + porter stemmer. ✗
- ⑩ [A] CFG + Features Structure. ✓
- ⑪ [C] Remove all PP production $NP \rightarrow NP PP$. ✗
- ⑫ [A] The list of authorized employees. ✗
- ⑬ [C] The verb violates its subcategorization frame. ✓
- ⑭ [B] Left recursion causing infinite recursion. ✓
- ⑮ [D] Dynamic chart construction combining predictor, scanner & completer. ✓
- ⑯ [A] Number feature conflict b/w subject & verb. ✓
- ⑰ [A] Tree 1 because it has higher probability. ✓
- ⑱ [A] Highly recursive grammar with unrestricted productions. ✗
- ⑲ [D] Earley parsing handles efficiency, while PCFG wants alternate. ✓
- ⑳ [C] CFG Parsing $\rightarrow$ Subcategorization checking $\rightarrow$ Feature unification $\rightarrow$ PCFG based. ✓
- ㉑ [C] Introduce a compound noun production $N \rightarrow \text{"living room"} N$. ✓
- ㉒ [A] Main subject noun phrase. ✓
- ㉓ [D] Verb subcategorization frame. ✓

- (24) [A] Shift reduce conflict $\times$
- (25) [B] Scanner $\times$
- (26) [B] Max likelihood estimation $\times$
- (27) [A] Tree A because it has the highest probability ✓
- (28) [A] Context free grammar only $\times$
- (29) [B] Recursive Top-Down Parser ✓
- (30) [A] Max Frequency matching $\times$
- (31) [A] $S \rightarrow NP VP \rightarrow Det N VP \rightarrow the robot VP \rightarrow V NP$ ✓
- (32) [C] The engineers develop the software ✓
- (33) [C] she gave the customer a refund ✓
- (34) [D] Expand S into NPVP ✓
- (35) [C] Completer $\times$
- (36) [B] Unification fails feature ✓
- (37) [A] The parse with the highest ✓
- (38) [B] Recursive Decent Parsing $\times$
- (39) [A] Parse A because it satisfies the constraints & has the highest probability ✓
- (40) [B] Reject the parse because grammatical agreement constraint ✓
- (41) [B] $NP \rightarrow Det N$ $\times$
- (42) [B] VP ✓
- (43) [A] Predictor $\times$
- (44) [B] Features automatically converted to plural ✓
- (45) [A] the max number of production rules ✓
- (46) [A] CFG supports recursive noun phrase generation using $NP \rightarrow NP$ ✓
- (47) [C] Structural (syntactic) ambiguity ✓
- (48) [C] Recursive grammar expansion ✓
- (49) [A] The verb "gave" requires additional complements ✓
- (50) [C] V NP PP ✓